Variation of parameters for higher order equations

Just like with undetermined coefficients, we're trying to find the solution to the nth order nonhomogeneous linear equation

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \ldots + p_1(t)y' + p_0(t)y = g(t)$$

by first finding the complementary solution (the solution to the associated homogeneous equation), and then using the method of variation of parameters to find the particular solution.

Remember that when we used variation of parameters with second order equations, we would solve the system of linear equations

$$u_1'y_1 + u_2'y_2 = 0$$

$$u_1'y_1' + u_2'y_2' = g(t)$$

because there were only two solutions in the solution set $\{y_1, y_2\}$, and so there were only two unknowns, u_1' and u_2'.

But with higher order equations, of course the solution set will be larger, for example $\{y_1, y_2, y_3, y_4, \ldots, y_n\}$. If we had a fourth order linear differential equation, for example, we'd have to solve this system:

$$u_1'y_1 + u_2'y_2 + u_3'y_3 + u_4'y_4 = 0$$

$$u_1'y_1' + u_2'y_2' + u_3'y_3' + u_4'y_4' = 0$$

$$u_1'y_1'' + u_2'y_2'' + u_3'y_3'' + u_4'y_4'' = 0$$

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$$u_1'y_1''' + u_2'y_2''' + u_3'y_3''' + u_4'y_4''' = g(t)$$

because we need the same number of equations in the system as we have solutions in the solution set. The forcing function $g(t)$ from the original differential equation is always the right side of the last equation in the system; each of the other equations will have 0 as its right side.

We can certainly solve the system using simple substitution and elimination techniques, but the larger the size of the solution set, and therefore the larger the size of the system of linear equations, the more convenient it becomes to use Cramer's Rule to solve the system.

Cramer's Rule

We'll always start by changing the nonhomogeneous equation to a homogeneous equation so that we can find the complementary solution, and then pull the fundamental set of solutions from that complementary solution.

We'll then use that fundamental set of solutions to build a set of Wronskian determinants. To illustrate what these determinants look like, let's assume that we're working with a fundamental set of four solutions, $\{y_1, y_2, y_3, y_4\}$. Then Cramer's Rule tells us that we'll need to find

$$W = \begin{vmatrix} y_1 & y_2 & y_3 & y_4 \\ y_1' & y_2' & y_3' & y_4' \\ y_1'' & y_2'' & y_3'' & y_4'' \\ y_1''' & y_2''' & y_3''' & y_4''' \end{vmatrix} \quad W_1 = \begin{vmatrix} 0 & y_2 & y_3 & y_4 \\ 0 & y_2' & y_3' & y_4' \\ 0 & y_2'' & y_3'' & y_4'' \\ 1 & y_2''' & y_3''' & y_4''' \end{vmatrix} \quad W_2 = \begin{vmatrix} y_1 & 0 & y_3 & y_4 \\ y_1' & 0 & y_3' & y_4' \\ y_1'' & 0 & y_3'' & y_4'' \\ y_1''' & 1 & y_3''' & y_4''' \end{vmatrix}$$

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$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 & y_4 \\ y_1' & y_2' & 0 & y_4' \\ y_1'' & y_2'' & 0 & y_4'' \\ y_1''' & y_2''' & 1 & y_4''' \end{vmatrix} \qquad W_4 = \begin{vmatrix} y_1 & y_2 & y_3 & 0 \\ y_1' & y_2' & y_3' & 0 \\ y_1'' & y_2'' & y_3'' & 0 \\ y_1''' & y_2''' & y_3''' & 1 \end{vmatrix}$$

Notice how W includes only the four solutions and their derivatives, while W_1, W_2, W_3, and W_4 each replace one column of W with a column full of zeros, ending with a 1 at the bottom of the column.

Then, instead of solving the system,

$$u_1'y_1 + u_2'y_2 + u_3'y_3 + u_4'y_4 = 0$$

$$u_1'y_1' + u_2'y_2' + u_3'y_3' + u_4'y_4' = 0$$

$$u_1'y_1'' + u_2'y_2'' + u_3'y_3'' + u_4'y_4'' = 0$$

$$u_1'y_1''' + u_2'y_2''' + u_3'y_3''' + u_4'y_4''' = g(t)$$

Cramer's Rule tells us that we can find the unknowns u_1', u_2', u_3', and u_4' by substituting the Wronskian determinants we found earlier into these equations:

$$u_1' = \frac{g(t)W_1}{W} \qquad u_2' = \frac{g(t)W_2}{W} \qquad u_3' = \frac{g(t)W_3}{W} \qquad u_4' = \frac{g(t)W_4}{W}$$

$$u_1 = \int \frac{g(t)W_1}{W}\,dt \quad u_2 = \int \frac{g(t)W_2}{W}\,dt \quad u_3 = \int \frac{g(t)W_3}{W}\,dt \quad u_4 = \int \frac{g(t)W_4}{W}\,dt$$

With these u_1, u_2, u_3, and u_4 values, the particular solution can be written as

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3 + u_4y_4$$

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$$y_p(t) = y_1 \int \frac{g(t)W_1}{W}\,dt + y_2 \int \frac{g(t)W_2}{W}\,dt + y_3 \int \frac{g(t)W_3}{W}\,dt + y_4 \int \frac{g(t)W_4}{W}\,dt$$

Remember that we can modify the system of linear equations and these formulas for the complementary and particular solutions depending on how many solutions we have in the fundamental set of solutions.

Let's try an example where we use Cramer's Rule, instead of solving the system of equations, to find the particular solution. We'll start with an

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Example

Use variation of parameters to find the general solution to the differential equation.

$$y''' - 6y'' + 12y' - 8y = 3e^{2t} + 4e^{3t}$$

This is the same example we worked in the last lesson using the method of undetermined coefficients. So we already know that the characteristic equation we get from the homogeneous equation is

$$r^3 - 6r^2 + 12r - 8 = 0$$

$$(r-2)(r-2)(r-2) = 0$$

and therefore that the complementary solution is

$$y_c(t) = c_1e^{2t} + c_2te^{2t} + c_3t^2e^{2t}$$

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From the complementary solution, we get the fundamental set of solutions $\{e^{2t}, te^{2t}, t^2e^{2t}\}$. With three solutions in the fundamental set, we'll build the following Wronskian determinants.

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{2t} & te^{2t} & t^2e^{2t} \\ 2e^{2t} & e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t} & 4e^{2t}+4te^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

$$= e^{2t}\begin{vmatrix} e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t}+4te^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

$$-te^{2t}\begin{vmatrix} 2e^{2t} & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix} + t^2e^{2t}\begin{vmatrix} 2e^{2t} & e^{2t}+2te^{2t} \\ 4e^{2t} & 4e^{2t}+4te^{2t} \end{vmatrix}$$

$$= e^{2t}[(e^{2t}+2te^{2t})(2e^{2t}+8te^{2t}+4t^2e^{2t}) - (2te^{2t}+2t^2e^{2t})(4e^{2t}+4te^{2t})]$$

$$-te^{2t}[(2e^{2t})(2e^{2t}+8te^{2t}+4t^2e^{2t}) - (2te^{2t}+2t^2e^{2t})(4e^{2t})]$$

$$+t^2e^{2t}[(2e^{2t})(4e^{2t}+4te^{2t}) - (e^{2t}+2te^{2t})(4e^{2t})]$$

$$= e^{2t}(2e^{4t}+4te^{4t}+4t^2e^{4t}) - te^{2t}(4e^{4t}+8te^{4t}) + t^2e^{2t}(4e^{4t})$$

$$= 2e^{6t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & te^{2t} & t^2e^{2t} \\ 0 & e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \\ 1 & 4e^{2t}+4te^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

$$= 0\begin{vmatrix} e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t}+4te^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

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$$-0\begin{vmatrix} te^{2t} & t^2e^{2t} \\ 4e^{2t}+4te^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix} + 1\begin{vmatrix} te^{2t} & t^2e^{2t} \\ e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \end{vmatrix}$$

$$= \begin{vmatrix} te^{2t} & t^2e^{2t} \\ e^{2t}+2te^{2t} & 2te^{2t}+2t^2e^{2t} \end{vmatrix}$$

$$= (te^{2t})(2te^{2t}+2t^2e^{2t}) - (t^2e^{2t})(e^{2t}+2te^{2t})$$

$$= t^2e^{4t}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{2t} & 0 & t^2e^{2t} \\ 2e^{2t} & 0 & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t} & 1 & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

$$= -0\begin{vmatrix} 2e^{2t} & 2te^{2t}+2t^2e^{2t} \\ 4e^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix}$$

$$+0\begin{vmatrix} e^{2t} & t^2e^{2t} \\ 4e^{2t} & 2e^{2t}+8te^{2t}+4t^2e^{2t} \end{vmatrix} - 1\begin{vmatrix} e^{2t} & t^2e^{2t} \\ 2e^{2t} & 2te^{2t}+2t^2e^{2t} \end{vmatrix}$$

$$= -\begin{vmatrix} e^{2t} & t^2e^{2t} \\ 2e^{2t} & 2te^{2t}+2t^2e^{2t} \end{vmatrix}$$

$$= -(e^{2t})(2te^{2t}+2t^2e^{2t}) + (t^2e^{2t})(2e^{2t})$$

$$= -2te^{4t}$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{2t} & te^{2t} & 0 \\ 2e^{2t} & e^{2t}+2te^{2t} & 0 \\ 4e^{2t} & 4e^{2t}+4te^{2t} & 1 \end{vmatrix}$$

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$$= 0\begin{vmatrix} 2e^{2t} & e^{2t}+2te^{2t} \\ 4e^{2t} & 4e^{2t}+4te^{2t} \end{vmatrix} - 0\begin{vmatrix} e^{2t} & te^{2t} \\ 4e^{2t} & 4e^{2t}+4te^{2t} \end{vmatrix} + 1\begin{vmatrix} e^{2t} & te^{2t} \\ 2e^{2t} & e^{2t}+2te^{2t} \end{vmatrix}$$

$$= \begin{vmatrix} e^{2t} & te^{2t} \\ 2e^{2t} & e^{2t}+2te^{2t} \end{vmatrix}$$

$$= (e^{2t})(e^{2t}+2te^{2t}) - (te^{2t})(2e^{2t})$$

$$= e^{4t}$$

Then the unknowns are

$$u_1' = \frac{g(t)W_1}{W} = \frac{(3e^{2t}+4e^{3t})(t^2e^{4t})}{2e^{6t}} = \frac{3t^2e^{6t}+4t^2e^{7t}}{2e^{6t}} = \frac{3t^2+4t^2e^t}{2}$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{(3e^{2t}+4e^{3t})(-2te^{4t})}{2e^{6t}} = -\frac{6te^{6t}+8te^{7t}}{2e^{6t}} = -3t-4te^t$$

$$u_3' = \frac{g(t)W_3}{W} = \frac{(3e^{2t}+4e^{3t})(e^{4t})}{2e^{6t}} = \frac{3e^{6t}+4e^{7t}}{2e^{6t}} = \frac{3+4e^t}{2}$$

and therefore

$$u_1 = \int \frac{g(t)W_1}{W}\,dt = \int \frac{3t^2+4t^2e^t}{2}\,dt = \frac{1}{2}t^3+2t^2e^t-4te^t+4e^t$$

$$u_2 = \int \frac{g(t)W_2}{W}\,dt = \int -3t-4te^t\,dt = -\frac{3}{2}t^2-4te^t+4e^t$$

$$u_3 = \int \frac{g(t)W_3}{W}\,dt = \int \frac{3+4e^t}{2}\,dt = \frac{3}{2}t+2e^t$$

Then the particular solution is

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$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(\frac{1}{2}t^3 + 2t^2e^t - 4te^t + 4e^t\right)e^{2t} + \left(-\frac{3}{2}t^2 - 4te^t + 4e^t\right)te^{2t}$$

$$+\left(\frac{3}{2}t + 2e^t\right)t^2e^{2t}$$

$$y_p(t) = \frac{1}{2}t^3e^{2t} + 2t^2e^{3t} - 4te^{3t} + 4e^{3t} - \frac{3}{2}t^3e^{2t} - 4t^2e^{3t} + 4te^{3t} + \frac{3}{2}t^3e^{2t} + 2t^2e^{3t}$$

$$y_p(t) = \frac{1}{2}t^3e^{2t} + 4e^{3t}$$

Then the general solution is the sum of the complementary and particular solutions,

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{2t} + c_2te^{2t} + c_3t^2e^{2t} + \frac{1}{2}t^3e^{2t} + 4e^{3t}$$

This is the same result we found using the method of undetermined coefficients in the previous lesson.

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